MATH 345 Skeleton Notes
Unit 1: Vectors, matrices, and linear maps
Section 1.2: Matrices

Matrices can be viewed in two complementary ways: as arrays of data and as functions acting on vectors. We begin with basic matrix arithmetic, then use matrix-vector multiplication to describe linear maps.

Matrices as arrays of numbers

Definition: Matrices

An $m \times n$ $\boxed{\text{matrix}}$ A is a rectangular array of $m \cdot n$ real numbers arranged in m $\boxed{\text{horizontal rows}}$ and n $\boxed{\text{vertical columns}}$:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The $\boxed{i\text{-th row}}$ of A is

$$\begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix},$$

and the $\boxed{j\text{-th column}}$ of A is

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}.$$

The number a_{ij}, which is in the i-th row and j-th column of A, is the $\boxed{(i,j)\text{-entry}}$ of A, and we often write $A = [a_{ij}]$. We say A is an " $\boxed{m \text{ by } n}$ " matrix.

Activity

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & -8 & -13 \end{bmatrix}$$

A is a 2×3 matrix. Compute the following:

a_{11}

1

Activity

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & -8 & -13 \end{bmatrix}$$

A is a 2×3 matrix. Compute the following:

a_{12}

2

Activity

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & -8 & -13 \end{bmatrix}$$

A is a 2×3 matrix. Compute the following:

a_{21}

5

Activity

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & -8 & -13 \end{bmatrix}$$

A is a 2×3 matrix. Compute the following:

a_{23}

-13

Activity

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & -8 & -13 \end{bmatrix}$$

A is a 2×3 matrix. Compute the following:

a_{32}

No such entry exists.

Data matrices

An important application of matrices (but far from the only one!) is to record data. The following table gives a few examples.

- **Object:** Monochrome image
  Meaning of an entry: X_{ij} is the pixel value in row i and column j.

- **Object:** Rainfall data
  Meaning of an entry: A_{ij} is the rainfall at location j on day i.

- **Object:** Asset returns
  Meaning of an entry: R_{ij} is the return of asset i in period j.

- **Object:** Feature matrix
  Meaning of an entry: X_{ij} is the value of feature j for entity i.

Matrix operations

Definition: Equality of matrices

Two matrices A and B are $\boxed{\text{equal}}$ if they have the same size and all the corresponding entries are equal.

Activity

Suppose

$$A = \begin{bmatrix} 3 & y \\ 4 & -7 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 6+x & -2 \\ 4 & -7 \end{bmatrix}$$

and $A = B$. Find x and y.

Since $A = B$, all entries of A must equal the corresponding entries in B. So it must be true, comparing corresponding entries, that

$$\begin{aligned} 3 &= 6 + x \\ y &= -2 \end{aligned}$$

Therefore, $x = -3$ and $y = -2$.

Definition: Sums of matrices

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are both $m \times n$ matrices, then their $\boxed{\text{sum}}$ $A + B$ is the matrix $C = [c_{ij}]$ where $c_{ij} = a_{ij} + b_{ij}$.

Activity

For the matrices

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & -8 & -13 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 3 & -4 \end{bmatrix}$$

calculate $A + B$.

$$\begin{aligned} A + B &= \begin{bmatrix} 1+0 & 2+2 & 3+1 \\ 5+1 & -8+3 & -13+(-4) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 4 & 4 \\ 6 & -5 & -17 \end{bmatrix}. \end{aligned}$$

Warning

For $A + B$ to be defined, A and B must be the same size. From now on, if we write $A + B$, assume that this is the case.

Definition: Scalar multiples of matrices

If $A = [a_{ij}]$ is an $m \times n$ matrix and r is a real number, then the scalar multiple of A by r, written rA, is the $m \times n$ matrix $C = [c_{ij}]$, where $c_{ij} = ra_{ij}$, that is, C is the matrix obtained by multiplying every entry of A by r.

Activity

For A as in the activity, calculate $-2A$.

$$-2A = \begin{bmatrix} -2 & -4 & -6 \\ -10 & 16 & 26 \end{bmatrix}$$

Definition: Linear combinations of matrices

If $A_1, A_2, \dots, A_k$ are $m \times n$ matrices and $c_1, c_2, \dots, c_k$ are real numbers, then an expression of the form

$$c_1 A_1 + c_2 A_2 + \cdots + c_k A_k$$

is called a

linear combination of $A_1, A_2, \dots, A_k$. The scalars $c_1, \dots, c_k$ are called the

coefficients of the linear combination.

This is the same idea as a linear combination of vectors from Vectors. The only difference is that the objects being combined are matrices of the same size.

Activity

Compute the following linear combination of matrices:

$$4 \begin{bmatrix} 0 & 2 \\ -3 & 3 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 4 & 2 \\ 6 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 8 \\ -12 & 12 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} \qquad = \begin{bmatrix} -2 & 7 \\ -15 & 11 \end{bmatrix}$$

Theorem

Let A, B, and C be $m \times n$ matrices.

1. $A + B = B + A$, i.e., matrix addition is $\boxed{\text{commutative}}$.
2. $A + (B + C) = (A + B) + C$, i.e., matrix addition is $\boxed{\text{associative}}$.
3. There is a unique $m \times n$ matrix O such that $A + O = A$ for any $m \times n$ matrix A. The matrix O is called the $m \times n$ $\boxed{\text{zero matrix}}$, and is the matrix with zeros in every entry.
4. For each $m \times n$ matrix A, there is a unique $m \times n$ matrix D such that $A + D = O$. The matrix D must be the matrix $-A = (-1)A$. The matrix $-A$ is called the $\boxed{\text{negative}}$ of A.

Let r and s be real numbers. Then

1. $r(sA) = (rs)A$.
2. $(r + s)A = rA + sA$.
3. $r(A + B) = rA + rB$.

Proof

We prove Property 1 only, i.e., the commutativity of addition. Let $A = [a_{ij}]$ and $B = [b_{ij}]$. Then:

$$\begin{aligned} A + B &= [a_{ij} + b_{ij}] \\ &= [b_{ij} + a_{ij}] \qquad \text{(since real numbers are commutative)} \\ &= B + A \end{aligned}$$

Definition

If $A = [a_{ij}]$ is an $m \times n$ matrix, then the $\boxed{\text{transpose}}$ of A, denoted $A^T = [a^T_{ij}]$, is the $n \times m$ matrix defined by

$$a^T_{ij} = a_{ji}$$

In other words, the transpose of A is obtained by *interchanging* the rows and the columns of A.

Activity

Compute the transpose for each of the given matrices:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & -8 & -13 \end{bmatrix}.$$

$$A^T = \begin{bmatrix} 1 & 5 \\ 2 & -8 \\ 3 & -13 \end{bmatrix}$$

Activity

Compute the transpose for each of the given matrices:

$$B = \begin{bmatrix} 5 & 2 & 3 \\ 6 & 2 & 3 \\ -1 & -2 & 3 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 5 & 6 & -1 \\ 2 & 2 & -2 \\ 3 & 3 & 3 \end{bmatrix}$$

Activity

Compute the transpose for each of the given matrices:

$$C = \begin{bmatrix} 10 \\ 20 \\ 30 \end{bmatrix}.$$

$$C^T = \begin{bmatrix} 10 & 20 & 30 \end{bmatrix}$$

Note

Observe from the previous activity that, when transposed, a column vector becomes a row vector. And vice versa.

Definition: Main diagonal

If $A = [a_{ij}]$ is an $m \times n$ matrix, the elements $a_{11}, a_{22}, a_{33}, \ldots$ are called the main diagonal of A. A matrix A is called diagonal if its only nonzero entries occur on its main diagonal.

Below are four matrices of various dimensions, with the main diagonal written in bold font.

$$\begin{bmatrix} \mathbf{a_{11}} & a_{12} \\ a_{21} & \mathbf{a_{22}} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{a_{11}} & a_{12} & a_{13} \\ a_{21} & \mathbf{a_{22}} & a_{23} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{a_{11}} & a_{12} & a_{13} \\ a_{21} & \mathbf{a_{22}} & a_{23} \\ a_{31} & a_{32} & \mathbf{a_{33}} \end{bmatrix}$$

$$\begin{bmatrix} \mathbf{a_{11}} \\ a_{21} \end{bmatrix}$$

Forming the transpose of a matrix A can be viewed as *flipping* A about its main diagonal.

Theorem

If r is a scalar and A and B are matrices of the appropriate sizes, then:

1. $(A^T)^T = A$.
2. $(A+B)^T = A^T + B^T$.
3. $(rA)^T = rA^T$.

Proof

Proof of property 2:
Let $A = [a_{ij}]$ and $B = [b_{ij}]$. Then $A + B = [c_{ij}]$ where $c_{ij} = a_{ij} + b_{ij}$.
Then

$$\begin{aligned}(A+B)^T &= [c_{ij}^T] \\ &= [c_{ji}] && \text{By definition of transpose} \\ &= [a_{ji} + b_{ji}] && \text{Since } c_{ij} = a_{ij} + b_{ij} \\ &= [a_{ji}] + [b_{ji}] && \text{By definition of matrix addition} \\ &= A^T + B^T && \text{By definition of transpose}\end{aligned}$$

Therefore, $(A+B)^T = A^T + B^T$.

Definition: Symmetry and skew-symmetry

A matrix A with real entries is called:

- Symmetric if $A^T = A$.
- Skew-symmetric if $A^T = -A$.

Warning

The previous definition only makes sense if the matrix A is square, i.e., if it has the same number of rows and columns.

Activity

Determine whether the following matrices are symmetric, skew symmetric, or neither:

$$A = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & -5 & 0 \end{bmatrix}$$

A is skew symmetric since $A^T = -A$.

Activity

Determine whether the following matrices are symmetric, skew symmetric, or neither:

$$B = \begin{bmatrix} 3 & 5 & 2 \\ 5 & 1 & 4 \\ 2 & 4 & -1 \end{bmatrix}$$

B is symmetric since $B^T = B$.

Activity

Determine whether the following matrices are symmetric, skew symmetric, or neither:

$$C = \begin{bmatrix} 1 & 2 & -3 \\ -2 & 0 & 5 \\ 3 & 5 & 0 \end{bmatrix}$$

C is neither symmetric nor skew symmetric.

Activity

Determine whether the following matrices are symmetric, skew symmetric, or neither:

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

D is both symmetric and skew symmetric.

Note

If a matrix A is both symmetric *and* skew symmetric, then A must be a zero matrix!

Note: Shape habit

Before adding or transposing matrices, first identify their shapes. A 2×3 matrix has two rows and three columns. Matrix addition requires the same shape, while transposing a 2×3 matrix produces a 3×2 matrix.